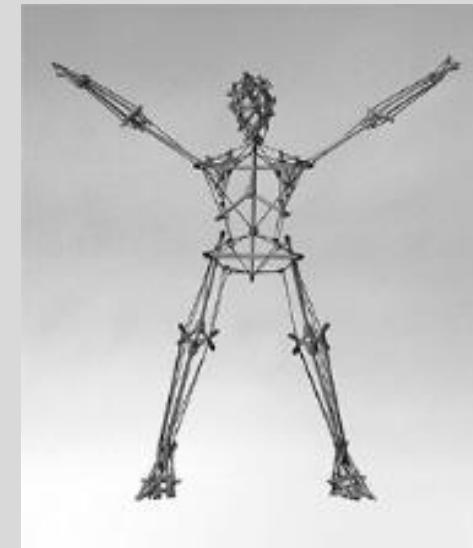
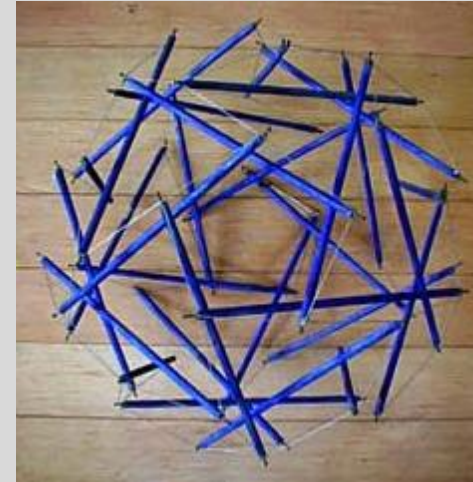


Lab Session Objectives

1. Students will understand the application and approach to evaluate a patient using the Area of Greatest Restriction (AGR).
2. Students will be able to perform screening finding the AGR for the upper half and lower half of the body.
3. Students will apply osteopathic treatment to the regions of AGR and understand how to sequence their treatment approach.
4. Student will review the history and principles of FPR.
5. Students will learn to diagnose and treat cervical, thoracic and lumbar SD with FPR.

Concept of Tensegrity in OMM

- Tensegrity = structural integrity via balance of **tension** and **compression**.
- **Body as a biotensegrity system:**
 - Rods = bones (compression).
 - Wires = muscles, fascia, ligaments (tension).
- Movement or dysfunction in one area changes forces throughout the body → informs whole-person assessment.



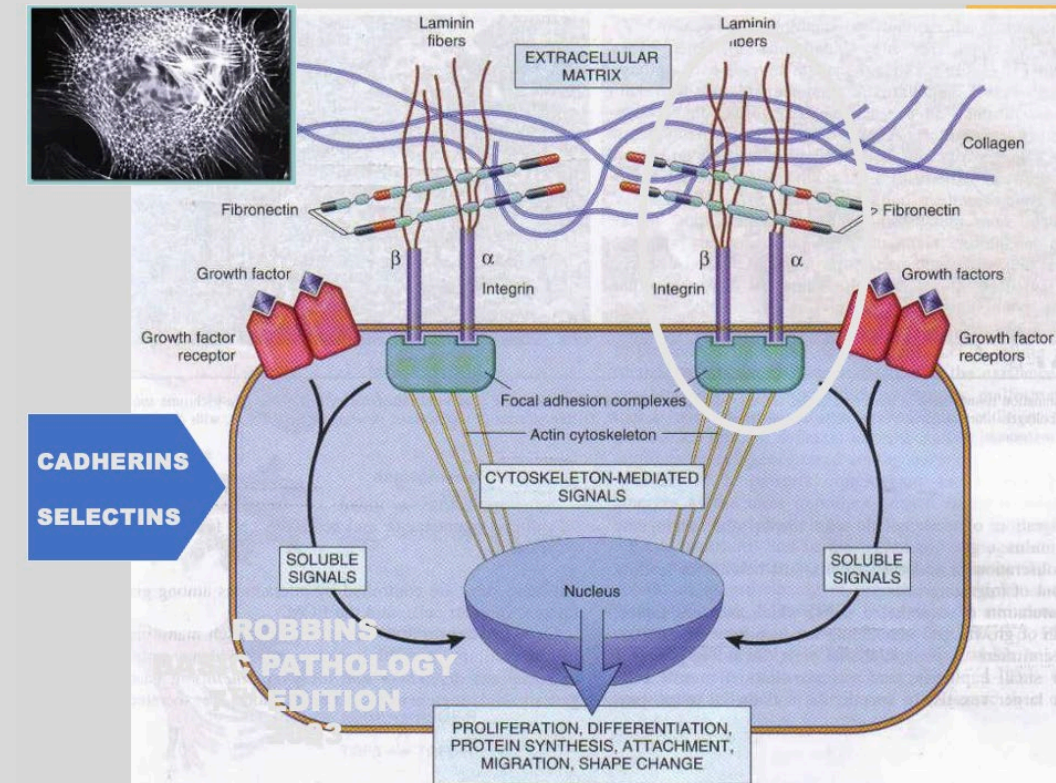
Mechanobiology and Diseases of Mechanotransduction

ANNALS OF MEDICINE 2003, DONALD E. INGBER, MD, PhD, HARVARD MEDICAL SCHOOL

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

- Mechanical forces are critical regulators of cellular biochemistry and gene expression as well as tissue development.
- **There is an “undeniable physical basis of disease.”**
- **Physical interventions can influence cell and tissue function.**
- Mechanical forces serve as important regulators at the cell and molecular levels, and they are equally potent as chemical cues.
- **“Forces that are applied to the entire organism (e.g., due to gravity or movement) would be distributed to individual cells via their adhesions to the ECM support scaffolds that link cells and tissues throughout the body.”**



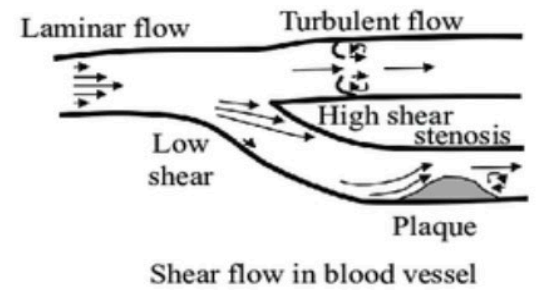
Mechanobiology and Diseases of Mechanotransduction

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- Mechanical forces directly regulate the shape and function of essentially all cell types.
- The global shape of the cell dictates its behavior (e.g., growth versus differentiation or apoptosis), and these effects are mediated through tension- dependent changes in cytoskeletal structure and mechanics.
- Manual type therapies have known the therapeutic value because they alter mechanotransduction.
- Most of the clinical problems that bring a patient to the doctor's office result from changes in tissue structure and mechanics.
- Abnormal cell and tissue responses to mechanical stress may actively contribute to the development of many diseases and ailments.
- **Consequently it may be wise to look for a physical cause for disease.**

Additional examples of mechanical effects on the human body

- Increased bone formation vs cartilage to mechanical loading
- Increased skeletal mass to stretch
- Increased hormone secretion of atrial natriuretic factor by heart cells to stretch
- Stretching of lung tissue regulates the synthesis of extracellular matrix
- Exposure to microgravity is associated with atrophy in heart, muscle, and bone, which is also frequented in aging
- Shear flow in blood vessel influences physiology and pathophysiology of endothelial cells and blood cells
- Increased growth of skin, bone, fibroblast to stretch.
- Kidney, bladder cells, pulmonary epithelial cells changing in growth and secretions to stretch.
- Increased deposition of bone to mechanical stress

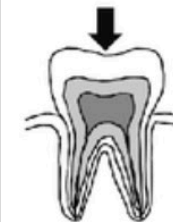


Contraction, compression, stretching

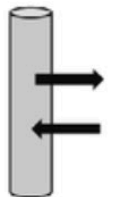


Stretching

Gravity



Compression



Hydrostatic, osmotic pressure

How does this information apply to Osteopathy?

- As osteopathic physicians we have the additional hands-on training to diagnose and treat physical dysfunctions that can influence these disease.
- Our treatment of the body has far reaching effects beyond just the musculoskeletal system but all the way down to the cellular level.
- You can treat many different disease processes by treating somatic dysfunction.

Area of Greatest Restriction & Sequencing

- Gives a starting place and an order to treatment.
- Allows the physician to focus in on the root of the problem and not just the areas of compensation.
- Reminds us to keep the whole person in mind during treatment and not to just focus on the area of complaint.

Application & Findings

Normal Joint

- You are at the **elastic/physiological barrier**.
- There will be a **yield, spring**, and “**soft end-feel**” with segmental motion testing.
- You can spring the joint simultaneously in three planes of motion

Restrictive Barrier

- When you challenge a dysfunctional area:
 - There will be **guarding** and **loss of tissue yield**.
 - **Joint spring is lost**, and you palpate a “**hard end-feel**.”
- **Find the most restricted/dysfunctional area.**
- **Grade dysfunctional regions: I, II, or III.**

Sequencing: Examination

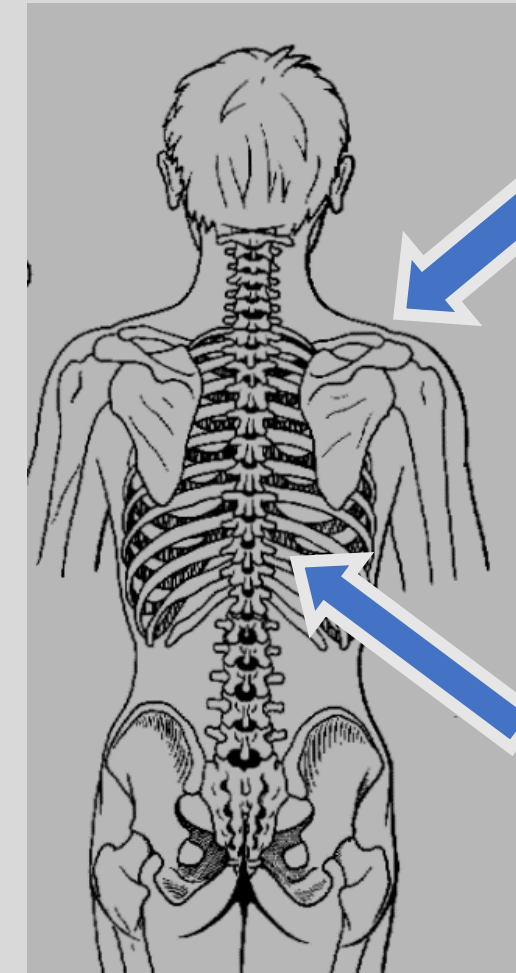
- Allow thumb to rest **lateral to spinous processes**, but **medial to transverse processes** — over the **facets** of all the cervical vertebrae.
- While in **neutral**, introduce **side bending** (*Type I*).
- Then slowly introduce **flexion/extension**, with each introduced movement until it segmentally moves down to the **base of the thumb** (*Type II*).
- Maintain this flexion. Now add **rotation** until movement segmentally moves down to the **base of the palpating thumb** (*Type II Enhanced*).
- **Findings:**
 - Normal: Soft & Springs
 - Dysfunctional: Hard & Lack Of Spring



Sequencing: examination

During The Palpatory Examination, Use Testing Movements As Follows:

- One Hand Introduces Movement.
- Other Hand Monitors The Segmental Response (“Listening Hand”)
 - Joint Mechanics
 - Parasegmental Tissue Response
- In Neutral, Introduce sidebending first, then flexion and extension, then rotation
- Slow “Sweeping” Movements Utilized
- Deliberate Motion: Interpreting The “End-Feel”
- Firm But Gentle Pressure
- Use the thumb or fingers on the listening hand to segmentally spring the dysfunctional area, in order to find the A.G.R. within the region. Identify the key within the dysfunctional area.
- Examine Both Standing & Sitting
(if key restriction is sacral - sitting → SI joint, if standing → gait dysfunction.)



OPERATOR'S
“MOTION
HAND”
(HAVE FINGER
TIPS OVER
CLAVICAL &
UPPER RIB
CAGE)

OPERATOR'S
PALPATING,
“LISTENING” &
“CHALLENGING
HAND”

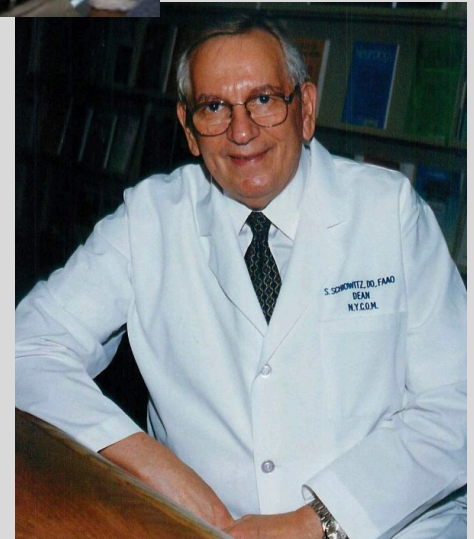
PRINCIPLE III
THORACIC, LUMBAR & SI / J

SEQUENCING: interpretation

- **Treatment Sequence**
- Once you find the **regional AGR**,
 - **Find the segmental AGR**
 - **Establish the segmental diagnosis**
 - **Start treatment there!**
- After treatment, **re-screen**.
- Then move to the **next regional AGR** and continue until the treated area releases **poorly** — **You're done for that visit.**

Facilitated Positional Release (FPR) Introduction

- Founder - Stanley Schiowitz, DO, FAAO
- Developed FPR in the 1990s
- Aimed to provide a rapid, effective method for relieving muscle tension and somatic dysfunction
- Influenced by earlier indirect techniques such as Strain-Counterstrain
- Designed for efficiency in clinical settings
- Commonly used in musculoskeletal medicine and pain management



Facilitated Positional Release (FPR)

Indications

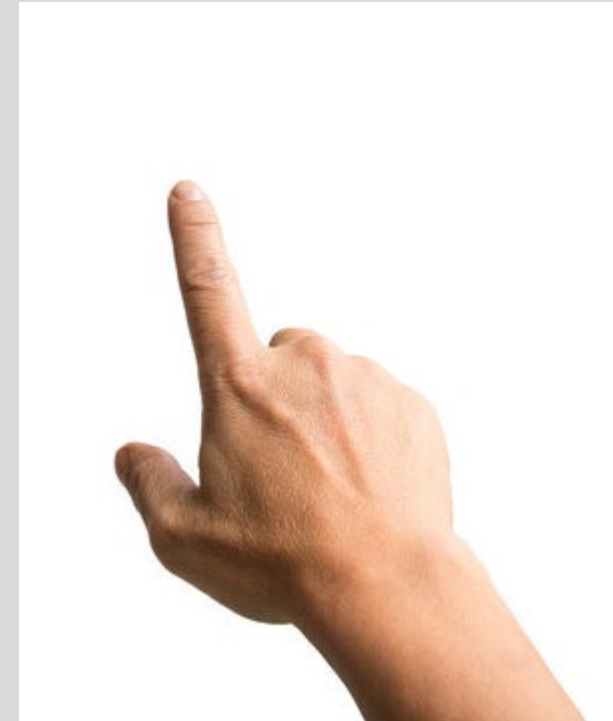
- Hypertonic muscle
- Somatic dysfunction (TART)

Contraindication

- Acute fractures or dislocations.
- Severe osteoporosis.
- Malignancy in the treatment area.
- Severe neurological deficits.
- Patient discomfort or inability to relax into the position.

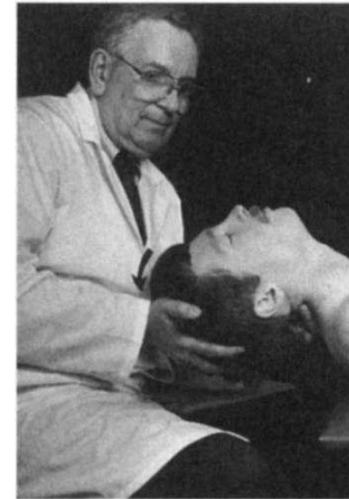
Key Steps for FPR

- Region of dysfunction placed in a neutral position
- Facilitating force applied (compression, torsion, or traction)
- Body part moved into position of ease
- Held for a few seconds (~3-5 seconds)
- Passive return to neutral
- Goal: Reduce muscle tension and improve range of motion
- Key is localization and being precise!
- Follow your monitoring finger!

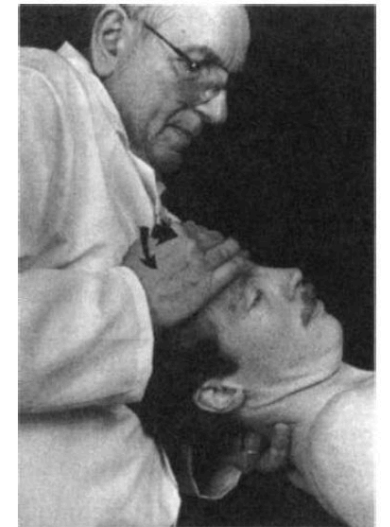


FPR - Cervical Spine – superficial muscle, Flexion/Extension SD

- **Positioning:** Patient supine, physician seated at head of table
- **Steps:**
 - Identify cervical segment with dysfunction
 - Neutralize region (reverse lordosis)
 - Apply axial compression
 - Move into position of ease (flexion/extension, side bending, rotation)
 - Hold 3-5 seconds, then release
 - Reassess



• **FIG. 29-2** Facilitated positional release treatment of muscle hypertonicity in the cervical region with extension and right side-bending added.

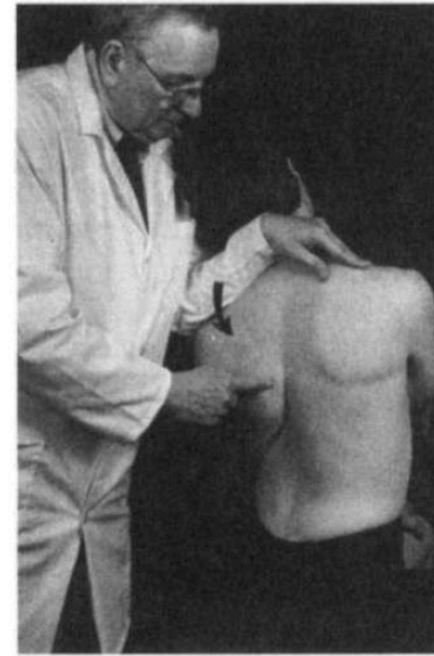


• **FIG. 29-3** Facilitated positional release treatment for C4 flexion dysfunction with right side-bending and rotation.

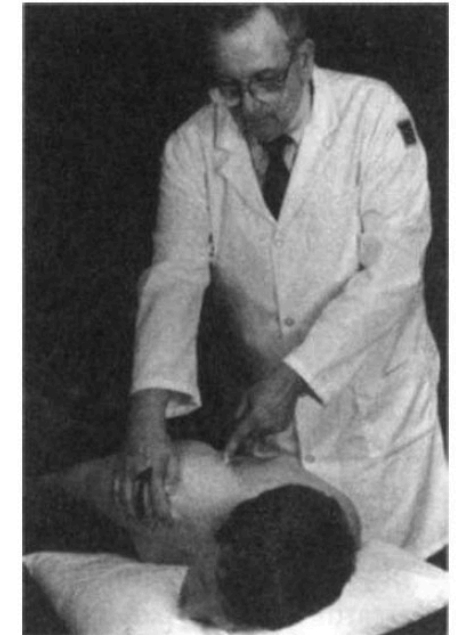
FPR - Thoracic Spine

- **Position:** Patient seated, physician standing on side of SD or on opposite side for prone position
- **Steps:**
 - Identify thoracic segment restriction
 - Adjust patient to neutralize spine
 - Apply gentle compression through shoulders to segment
 - Move into position of ease in all 3 planes
 - Hold 3-5 seconds, then release

DiGiovanna, E.L., Schiowitz, S. and Dowling, D.J. (2005) An Osteopathic Approach to Diagnosis and Treatment. 3rd Edition, Lippincott Williams & Wilkins, Philadelphia. Chapter 40



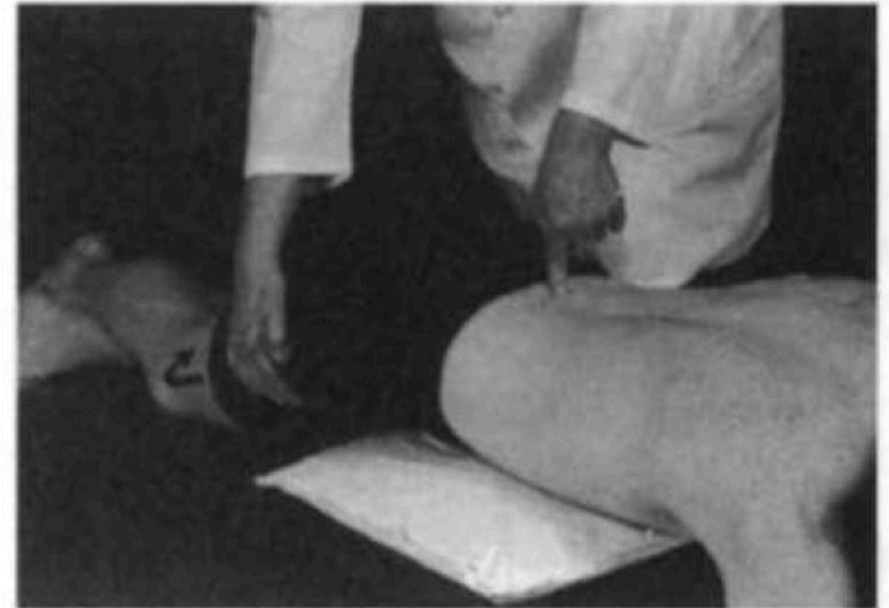
• **FIG. 40-2** Facilitated positional release treatment for thoracic superficial muscle hypertonicity; left side-bending added.



• **FIG. 40-4** A torsional motion is added, turning the shoulder upward and creating compression, side-bending, and rotation.

FPR - Lumbar Spine Muscle hypertonicity

- **Patient Position:** prone
- **Steps:**
 - Identify lumbar hypertonicity
 - Pillow to neutralize lumbar lordosis
 - Standing on opposite side bring patients legs towards affected side to sidebend the pelvis over your finger.
 - Cross far leg over ipsilateral ankle.
 - Extend contralateral thigh and externally rotate to monitoring finger.
 - Hold 3-5 seconds, then release.



■ **FIG. 51-1** Facilitated positional release treatment for the left low back, superficial muscle hypertonicity.

DiGiovanna, E.L., Schiowitz, S. and Dowling, D.J. (2005) An Osteopathic Approach to Diagnosis and Treatment. 3rd Edition, Lippincott Williams & Wilkins, Philadelphia. Chapter 51

FPR - Lumbar Spine Extension SD

- **Patient Position:** prone
- **Steps:**
 - Monitor lumbar Extension SD
 - Pillow to neutralize lumbar lordosis
 - Standing on opposite side and extend and internally rotate the hip till motion is felt
 - Cross far leg over ipsilateral ankle.
 - Extend leg slowly.
 - Hold 3-5 seconds, then release.



• FIG. 51-2 Facilitated positional release treatment for a lumbar extension somatic dysfunction.

DiGiovanna, E.L., Schiowitz, S. and Dowling, D.J. (2005) An Osteopathic Approach to Diagnosis and Treatment. 3rd Edition, Lippincott Williams & Wilkins, Philadelphia. Chapter 51

FPR - Lumbar Spine Flexion SD

- **Patient Position:** prone
- **Steps:**
 - Monitor lumbar Flexion SD
 - Pillow to neutralize lumbar lordosis
 - Sit same side and flex and adduct and internally rotate the hip till motion is felt
 - Add compression superiorly to monitoring hand for 3-5 seconds, then release.



• FIG. 51-4 Facilitated positional release treatment for a lumbar flexion somatic dysfunction.

Summary

- Osteopathic screening through finding the area of greatest restriction is one approach to help target key areas to start treatment.
- Utilizing a systematic approach will help to keep the whole person and body regions in mind and prevent focusing on the patient complaint.
- Utilization of finding the area of greatest restriction can help you find the true key problem that is causing problems in your patients.
- Understanding fascial connections will help to improve your understanding of treating the body.
- FPR is a safe and effective osteopathic treatment approach to address somatic dysfunctions of the spine